Can we improve tomorrow's hourly capacity plan?

Recommendation: pilot Ridge; retain seasonal naive as the fallback.

- Ridge MAE: 64.1 rentals/hour versus 71.9 for last week.

- Tuned boosting improves the baseline by 3.7%, but trails Ridge.

- Use forecasts to inform aggregate planning; require a prospective pilot before operations use.

One issuance. Twenty-four hourly forecasts.

Decision owner: a hypothetical bike-sharing operations manager.

- At 11:59 PM, predict the next calendar day's 24 system-wide rental counts.

- Current hypothetical process: repeat observed rentals from the same hour last week.

- Improved process: review hourly rental-throughput plans and staffing needs with uncertainty.

Real trips, with real limitations

Source: UCI Bike Sharing, Capital Bikeshare, Washington DC, 2011-2012.

- 17,379 observed hours; 165 missing slots stay missing.

- Counts measure completed rentals, not unserved customers or station inventories.

- Weather observations are excluded; no archived forecasts are available in the source dataset.

The forecast cannot see tomorrow

Availability checks and a chronological holdout protect the evaluation.

- Calendar plus lags 24 / 48 / 168 hours; rolling windows end 24 hours before each target.
- Fit through Oct 7, 2012; calibrate Oct 8-Nov 4; test Nov 5-Dec 30.
- 1,302 common scored hours; missing targets or baseline references excluded.
- Tune only on earlier time folds. Past test days refresh lags after observation.

The simpler regression wins on MAE

All three methods use the same held-out observations; lower is better.

Same hour last week: MAE 71.95 | RMSE 119.50

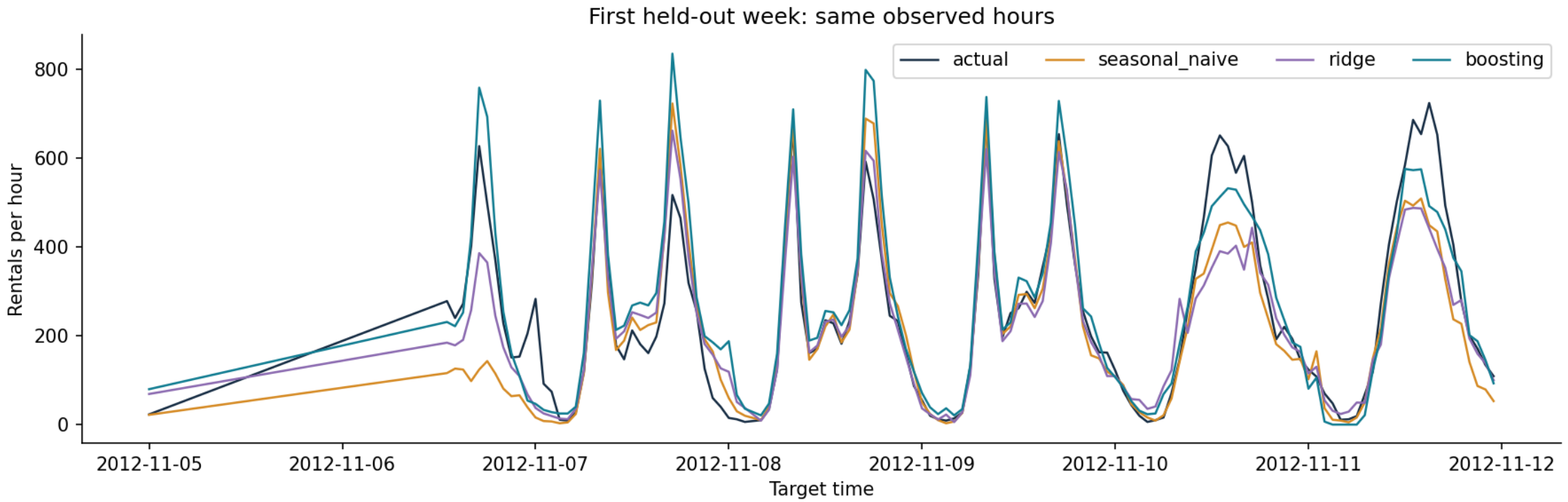
Ridge regression: MAE 64.10 | RMSE 96.10

Tuned boosting: MAE 69.30 | RMSE 108.42

Rental counts per hour. Model complexity did not produce the strongest result.

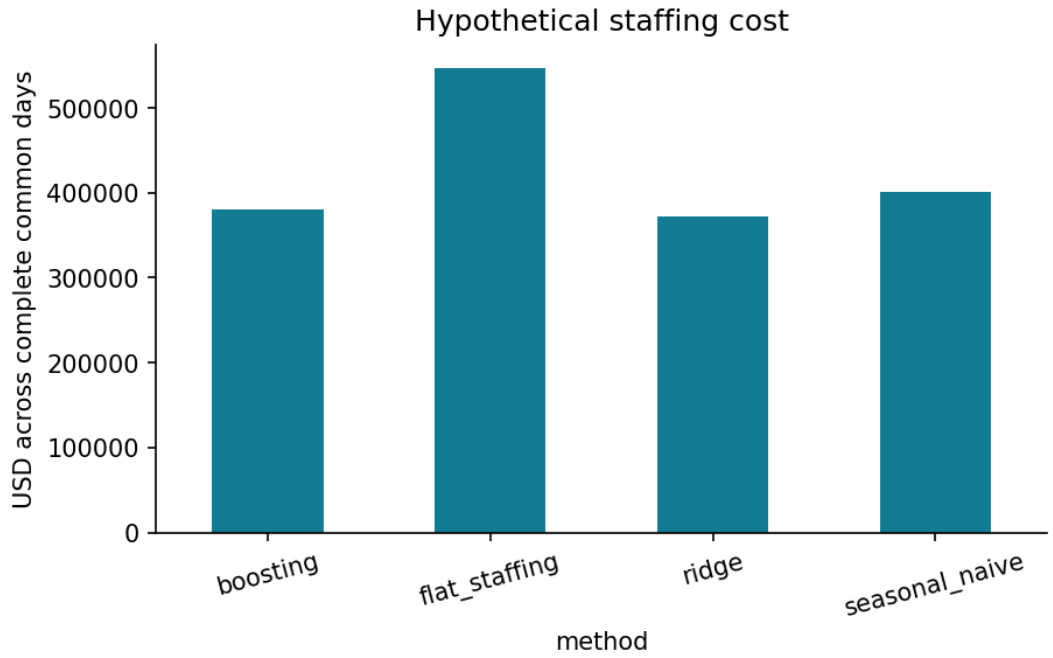
Where forecasts follow and miss the hourly pattern

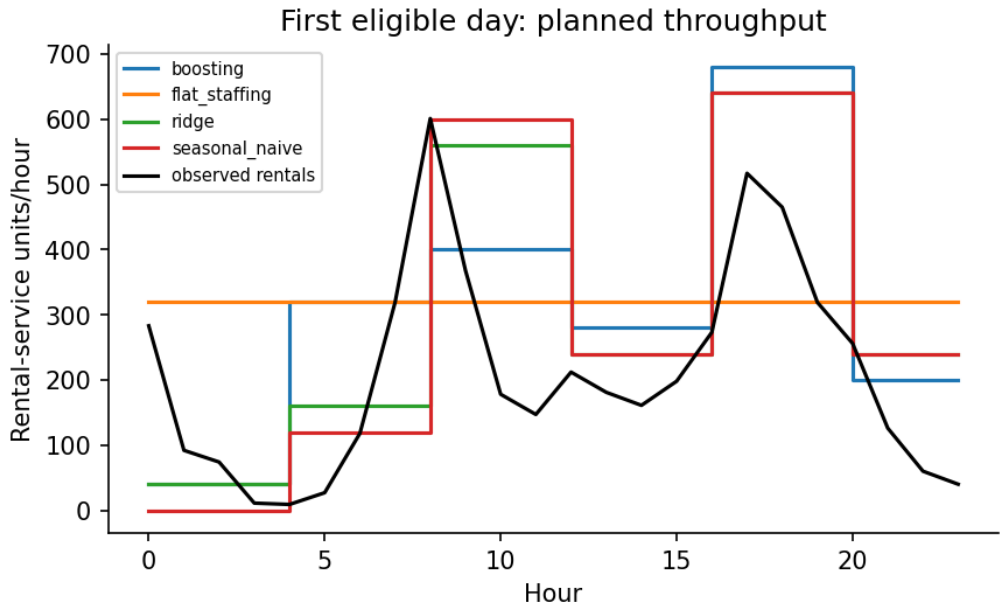
First held-out week, selected in advance; gaps are not invented observations.



From forecasts to shift assignments with Gurobi

Six 4-hour shifts; at most 48 worker-shift assignments/day; all operating quantities are assumptions.



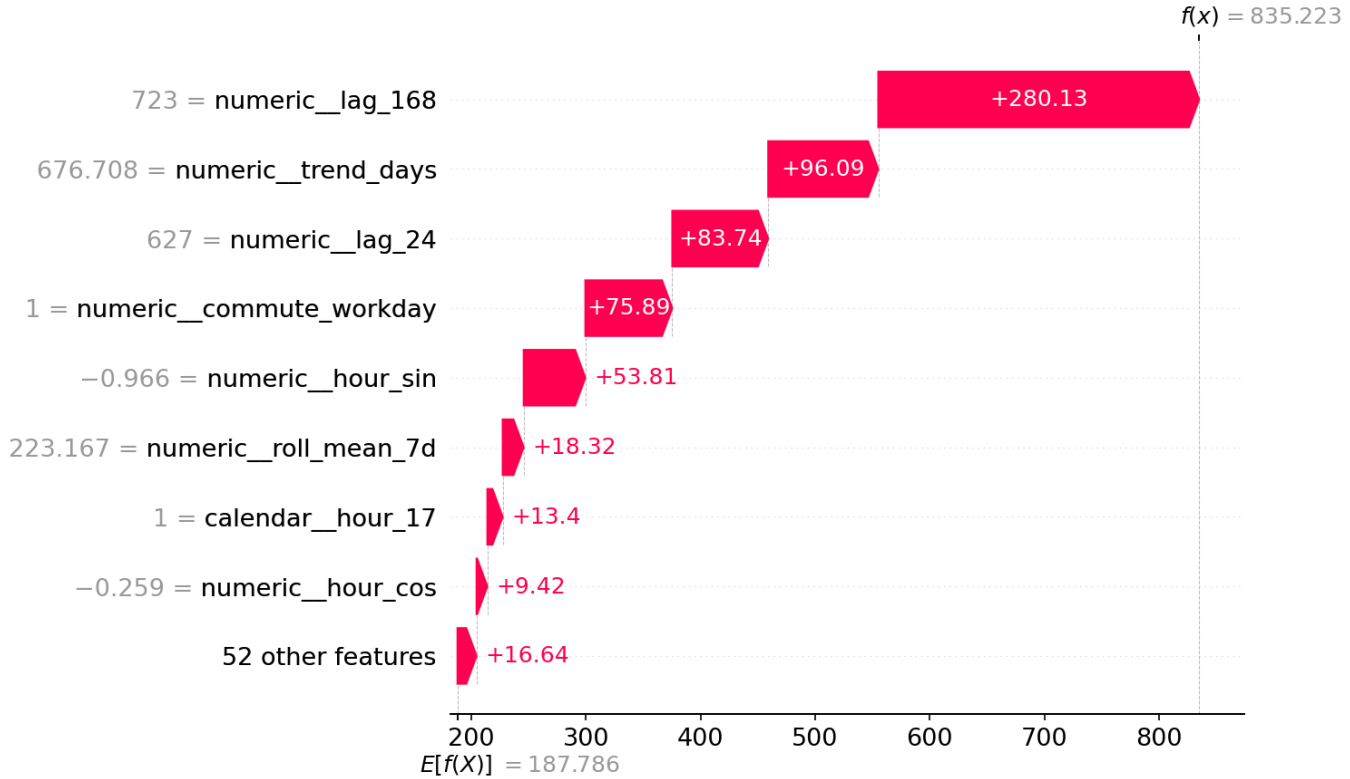


48 complete days: seasonal $401,486; Ridge $372,872; boosting $380,462.

144 certified optimal forecast-based solves; retrospective proxy costs, not realized savings.

Explanations help; intervals still miss their target

Tree SHAP attributes one prediction; it does not establish causality.

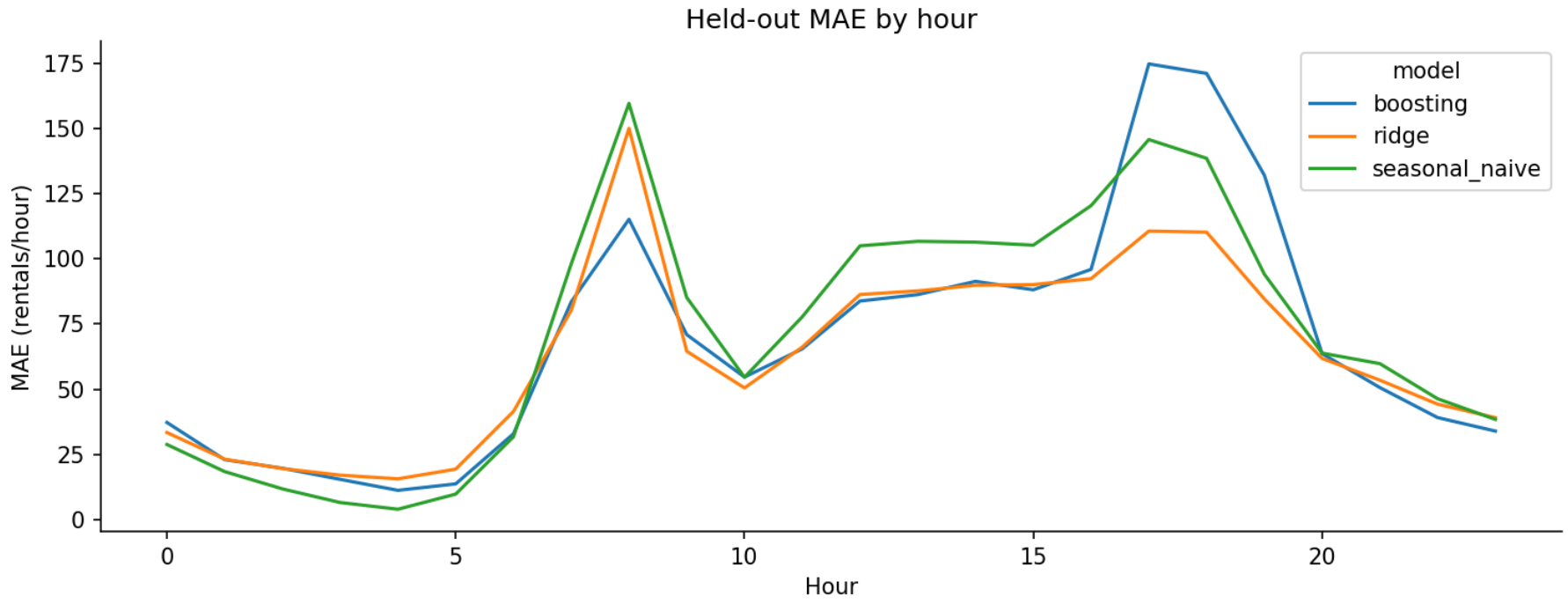


Nominal coverage: 90%. Observed coverage: 85.6%. Mean width: 251.4 rentals.

Temporal shift and dependence weaken exchangeability assumptions. Coverage requires monitoring.

Commute hours deserve closer review

Sliced errors identify stress points; only autumn and winter are represented in the test.



Worst boosting slice: hour=17; MAE 174.7, RMSE 210.1; n=55.

Pilot, monitor, then decide

Ridge is the current lead candidate; boosting is a documented challenger.

- Validate data latency, calendars, service constraints, and cost assumptions with operations.

- Monitor weekly error, missing history, peak-hour misses, interval coverage, and drift.

- Run a new prospective comparison before deployment; obtain station-level data for allocation.

- Validate forecasts prospectively before making operational commitments.